

Black Box Linear Algebra with the LinBox Library

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Overview

Preliminary Exam

- Preconditioners based on Beneš networks (Chapter 2)
- Preconditioners and determinantal divisors (Chapter 3)
- LinBox (Chapter 5)
- <http://www.math.ncsu.edu/~wjturner/2001/prelim.pdf>

Now

- Linearly generated matrix sequences and block Wiedemann method (Chapter 4)
- More LinBox (Chapter 5)

Block Wiedemann Method

Wiedemann method:

- $u, v \in \mathbb{F}^n$
- $\{u^\top A^i v\}_{i=0}^\infty \in \mathbb{F}^{\mathbb{Z}_{\geq 0}}$ linearly generated by f^A

Block Wiedemann method:

- $X \in \mathbb{F}^{n \times \beta_1}, Y \in \mathbb{F}^{n \times \beta_2}$
- $\{X^\top A^i Y\}_{i=0}^\infty \in (\mathbb{F}^{\beta_1 \times \beta_2})^{\mathbb{Z}_{\geq 0}}$

Linearly Generated Matrix Sequences

$\{B_i\}_{i=0}^{\infty} \in (\mathbb{F}^{\beta_1 \times \beta_2})^{\mathbb{Z}_{\geq 0}}$ is linearly generated by:

- $g = \sum_{i=0}^d g[i] \lambda^i \in \mathbb{F}[\lambda]$ if $g \neq 0$ and

$$\sum_{i=0}^d g[i] B_{i+j} = 0^{\beta_1 \times \beta_2}, \quad \forall j \geq 0.$$

- $C = \sum_{i=0}^d C[i] \lambda^i \in \mathbb{F}^{\beta_2}[\lambda]$ from the right if $C \neq 0$ and

$$\sum_{i=0}^d B_{i+j} C[i] = 0^{\beta_1}, \quad \forall j \geq 0.$$

- $G = \sum_{i=0}^d G[i] \lambda^i \in \mathbb{F}^{\beta_2 \times \beta_2}[\lambda]$ from the right if $\det(G) \neq 0$ and

$$\sum_{i=0}^d B_{i+j} G[i] = 0^{\beta_1 \times \beta_2}, \quad \forall j \geq 0.$$

$\{B_i\}_{i=0}^{\infty}$ linearly generated by polynomial if and only if generated from the right by matrix polynomial

$$g \in \mathbb{F}[\lambda] \text{ generates } \{B_i\}_{i=0}^{\infty}$$

$\Downarrow$

$$gI \text{ generates } \{B_i\}_{i=0}^{\infty} \text{ from the right}$$

and

$$G \in \mathbb{F}^{\beta_1 \times \beta_2}[\lambda] \text{ generates } \{B_i\}_{i=0}^{\infty} \text{ from the right}$$

$\Downarrow$

$$G \operatorname{adj}(G) = \det(G)I \text{ generates } \{B_i\}_{i=0}^{\infty} \text{ from the right}$$

$\Downarrow$

$$\det(G) \text{ generates } \{B_i\}_{i=0}^{\infty}$$

$\{B_i\}_{i=0}^{\infty}$ is linearly generated

Module of Right Generating Vectors

- Set $\mathbb{G}$ of right generating vectors of $\{B_i\}_{i=0}^{\infty}$ is a module over $\mathbb{F}[\lambda]$
- $\mathbb{G}$ has a basis over $\mathbb{F}[\lambda]$ of β_2 elements
- Let $b_1, \dots, b_{\beta_2}$ and $b'_1, \dots, b'_{\beta_2}$ be bases over $\mathbb{F}[\lambda]$ or $\mathbb{G}$.

$$G_1 = [b_1 \ \cdots \ b_{\beta_2}] \text{ and } G_2 = [b'_1 \ \cdots \ b'_{\beta_2}]$$
$$\Downarrow$$
$$\exists \text{ unimodular matrix } U \text{ such that } G_1 U = G_2$$

- Minimal generating matrix polynomial F for $\{B_i\}_{i=0}^{\infty}$ is Popov canonical form for equivalence.

Popov Form

$$M = \begin{bmatrix} M_1 & \cdots & M_{\beta_2} \end{bmatrix} \in \mathbb{F}^{\beta_2 \times \beta_2}[\lambda]$$

$$M_i \text{ is } i\text{-th column of } M; [M]_c = \begin{bmatrix} \text{lc}(M_1) & \cdots & \text{lc}(M_{\beta_2}) \end{bmatrix}$$

$$M \text{ is column reduced: } \text{rank}([M]_c) = \text{rank}(M)$$

1. the column degrees are increasingly ordered;
2. the last entry of degree $\deg(M_j)$ in M_j is monic and is called the pivot element of column j with row index r_j ;
3. if $\deg(M_j) = \deg(M_k)$ and $j < k$, then $r_j < r_k$;
4. all entries in a row containing a pivot element have degrees lower than that of the pivot element.

Properties of F

- $\text{rank}([F]_c) = \text{rank}(F) = \beta_2$
- $\gamma_2 = \deg(F) \leq \deg(f)$
- $\gamma_1 \leq \deg(f)$ is minimal integer such that

$$H(\gamma_1, \gamma_2 + 1) = \begin{bmatrix} B_0 & B_1 & \cdots & B_{\gamma_2} \\ B_1 & B_2 & \cdots & B_{\gamma_2+1} \\ \vdots & \vdots & \ddots & \vdots \\ B_{\gamma_1-1} & B_{\gamma_1} & \cdots & B_{\gamma_1+\gamma_2-1} \end{bmatrix}$$

has maximal rank

- $\nu_1 \geq \gamma_1, \nu_2 \geq \gamma_2 \Rightarrow \text{rank}(H(\nu_1, \nu_2 + 1)) = \text{rank}(H(\gamma_1, \gamma_2 + 1))$
- $\deg(\det(F)) \geq \text{rank}(H(\gamma_1, \gamma_2 + 1))$
- G generates $\{B_i\}_{i=0}^{\infty} \Rightarrow \exists M$ with $\det(M) \neq 0$ and $G = FM$
 - $\deg(\det(G)) \geq \deg(\det(F))$
 - $\deg(G) \geq \deg(F)$

$$\begin{aligned}
& C = \sum_{i=0}^d C[i] \lambda^i \text{ generates } \{B_i\}_{i=0}^{\infty} \\
& \quad \Updownarrow \\
& \sum_{j=0}^d B_{i+j} C[j] = 0^{\beta_1}, 0 \leq i \leq \nu_1 - 1 \\
& \quad \Updownarrow \\
& \left(\sum_{i=0}^{\nu_1 + \nu_2 - 1} B_i \lambda^i \right) \hat{C}(\lambda) \equiv C^{(\text{res})}(\lambda) \pmod{\lambda^{\nu_1 + \nu_2}} \\
& \quad \Updownarrow \\
& \left[I \quad - \left(\sum_{i=0}^{\nu_1 + \nu_2 - 1} B_i \lambda^i \right) \right] \begin{bmatrix} C^{(\text{res})} \\ \hat{C} \end{bmatrix} = \lambda^{\nu_1 + \nu_2} R(\lambda)
\end{aligned}$$

where $\nu_1 \geq \gamma_1$, $\nu_2 \geq d$, $\hat{C} = \text{rev}_d(C)$, and $C^{(\text{res})} \in \mathbb{F}^{\beta_1}[\lambda]$.

Vector Hermite-Padé Approximant

The polynomial tuple $\mathcal{P} = (\mathcal{P}_1, \dots, \mathcal{P}_m)$ is a vector Hermite-Padé approximant for the vector polynomials $\mathcal{G}_1, \dots, \mathcal{G}_m \in \mathbb{F}^s[\lambda]$ of type $(\mathcal{N}, \sigma)$ with the multiindex $\mathcal{N} = (\mathcal{N}_1, \dots, \mathcal{N}_m)$ if there exists a vector polynomial $R \in \mathbb{F}^s[\lambda]$ such that

$$\sum_{i=1}^m \mathcal{G}_i(\lambda) \mathcal{P}_i(\lambda) = \lambda^\sigma R(\lambda)$$

and $\deg(\mathcal{P}_i) \leq \mathcal{N}_i$ for $1 \leq i \leq m$

Power Hermite-Padé Approximant

$\mathcal{P}$ is a power Hermite-Padé approximant of the type $(\mathcal{N}, \sigma, s)$ given by the polynomial tuple $\mathcal{F} = (\mathcal{F}_1, \dots, \mathcal{F}_m)$ with the multiindex $\mathcal{N} = (\mathcal{N}_1, \dots, \mathcal{N}_m)$ if there exists a polynomial $R' \in \mathbb{F}[\lambda]$ such that

$$\mathcal{F}(\lambda) \cdot \mathcal{P}(\lambda^s) = \sum_{i=1}^m \mathcal{F}_i(\lambda) \mathcal{P}_i(\lambda^s) = \lambda^\sigma R'(\lambda)$$

and $\deg(\mathcal{P}_i) \leq \mathcal{N}_i$ for $1 \leq i \leq m$

Vector and Power Hermite-Padé Approximants

$$\sum_{i=1}^m \mathcal{G}_i(\lambda) \mathcal{P}_i(\lambda) = \lambda^\sigma R(\lambda)$$

$$\Leftrightarrow$$

$$\mathcal{F}(\lambda) \cdot \mathcal{P}(\lambda^s) = \sum_{i=1}^m \mathcal{F}_i(\lambda) \mathcal{P}_i(\lambda^s) = \lambda^{\sigma s} R'(\lambda) \in \mathbb{F}[\lambda]$$

where $\mathcal{F}_i(\lambda) = (1, \lambda, \lambda^2, \dots, \lambda^{s-1}) \cdot \mathcal{G}_i(\lambda^s)$

$C = \sum_{i=0}^d C[i] \lambda^i$ generates $\{B_i\}_{i=0}^{\infty}$

$$\begin{array}{c} \Updownarrow \\ \left[\mathcal{F}_i \cdots \mathcal{F}_{\beta_1+\beta_2} \right] \begin{bmatrix} C^{(\text{res})}(\lambda^{\beta_1}) \\ \hat{C}(\lambda^{\beta_1}) \end{bmatrix} = \lambda^{\nu_1+\nu_2} R'(\lambda) \in \mathbb{F}[\lambda] \end{array}$$

where $\mathcal{F}_i(\lambda) = (1, \lambda, \lambda^2, \dots, \lambda^{\beta_1-1}) \cdot \mathcal{G}_i(\lambda^s)$ and $\mathcal{G}_i$ is the i -th column of $\mathcal{G}(\lambda) = \begin{bmatrix} I & - \left(\sum_{i=0}^{\nu_1+\nu_2-1} B_i \lambda^i \right) \end{bmatrix}$

Defect

Let $\mathcal{P}$ be a solution to a power Hermite-Padé approximant problem of type $(\mathcal{N}, \sigma, s)$. Then, the defect of $\mathcal{P}$ is one more than the minimum difference in degree of a member polynomial and the corresponding degree bound,

$$\text{dct}(\mathcal{P}) = \min_{1 \leq i \leq \beta_1 + \beta_2} \{ \mathcal{N}_i + 1 - \deg(\mathcal{P}_i) \}$$

If

- $\{B_i\}_{i=0}^{\infty} \in (\mathbb{F}^{\beta_1 \times \beta_2})^{\mathbb{Z}_{\geq 0}}$ is linearly generated
- $\nu_1 \geq \gamma_1$ and $\nu_2 \geq \gamma_2$
- $\mathcal{P} = (\mathcal{P}_1, \dots, \mathcal{P}_{\beta_1 + \beta_2})$ is solution to PHPA $(\mathcal{N}, (\nu_1 + \nu_2)\beta_1, \beta_1)$
- $\mathcal{N}_i = \begin{cases} \nu_2 - 1 & \text{if } 1 \leq i \leq \beta_1, \\ \nu_2 & \text{if } \beta_1 + 1 \leq i \leq \beta_1 + \beta_2 \end{cases}$
- $\mathcal{F}_i(\lambda) = (1, \lambda, \lambda^2, \dots, \lambda^{\beta_1 - 1}) \cdot \mathcal{G}_i(\lambda^{\beta_1})$
- $\mathcal{G}_i$ is i -th column of $\mathcal{G}(\lambda) = [I - (\sum_{i=0}^{\nu_1 + \nu_2 - 1} B_i \lambda^i)]$

Then

- $\text{rev}_{\nu_2 + 1 - \text{dct}(\mathcal{P})} \left([\mathcal{P}_{\beta_1 + 1} \ \mathcal{P}_{\beta_1 + 2} \ \cdots \ \mathcal{P}_{\beta_1 + \beta_2}]^T \right) \in \mathbb{G}$

Fast Power Hermite-Padé Solver

Input: $m \geq 2, s \in \mathbb{Z}_{\geq 0}, \mathcal{F} = (\mathcal{F}_1, \dots, \mathcal{F}_m)$, multiindex $\mathcal{N} = (\mathcal{N}_1, \dots, \mathcal{N}_m)$

Output: $\{\mathcal{P}^{[i,\sigma]}\}_{i=1}^m$ and $\{\text{dct}(\mathcal{P}^{[i,\sigma]}) = d_{i,\sigma} + 1\}_{i=1}^m$ such that any solution $\mathcal{P}$ to the power Hermite-Padé approximant problem can be written as $\mathcal{P} = \sum_{i=1}^m g_i \mathcal{P}^{[i,\sigma]}$ where $\deg(g_i) < \text{dct}(\mathcal{P}^{[i]})$.

```

1: for  $1 \leq l \leq m$  do
2:    $d_{l,0} \leftarrow \mathcal{N}_l; \mathcal{P}^{[l,0]} \leftarrow e_l$  /*  $l$ -th unit vector */
3:   for  $0 \leq k \leq \sigma - 1$  do
4:     for  $1 \leq l \leq m$  do
5:        $c_{l,k} \leftarrow \lambda^{-k} \mathcal{P}^{[l,k]}(\lambda^s) \cdot \mathcal{F}(\lambda) \big|_{\lambda=0}$ 
6:        $\Lambda_k \leftarrow \{l \mid c_{l,k} \neq 0\}$ 
7:       if  $\Lambda_k = \{\}$  then
8:         for  $1 \leq l \leq m$  do
9:            $\mathcal{P}^{[l,k+1]} \leftarrow \mathcal{P}^{[k,l]}; d_{l,k+1} \leftarrow d_{l,k}$ 
10:      else
11:         $d_k \leftarrow \max\{d_{l,k} \mid l \in \Lambda_k\}$ 
12:         $\pi \leftarrow \pi_k \leftarrow \min\{j \mid d_{j,k} = d_k\}$ 
13:        for  $1 \leq l \leq m$  do
14:          if  $l \notin \Lambda_k$  then
15:             $\mathcal{P}^{[l,k+1]} \leftarrow \mathcal{P}^{[l,k]}; d_{l,k+1} \leftarrow d_{l,k}$ 
16:          else if  $l = \pi$  then /*  $l \in \Lambda_k$  */
17:             $\mathcal{P}^{[l,k+1]} \leftarrow \lambda \mathcal{P}^{[l,k]}; d_{l,k+1} \leftarrow d_{l,k} - 1$ 
18:          else /*  $l \in \Lambda_k, l \neq \pi$  */
19:             $\mathcal{P}^{[l,k+1]} \leftarrow \mathcal{P}^{[l,k]} - \frac{c_{l,k}}{c_{\pi,k}} \mathcal{P}^{[\pi,k]}; d_{l,k+1} \leftarrow d_{l,k}$ 

```


If

- $\{B_i\}_{i=0}^{\infty} \in (\mathbb{F}^{\beta_1 \times \beta_2})^{\mathbb{Z}_{\geq 0}}$ is linearly generated
- $\nu_1 \geq \gamma_1$ and $\nu_2 \geq \gamma_2$
- $\mathcal{P}[1, (\nu_1 + \nu_2)\beta_1], \dots, \mathcal{P}[\beta_1 + \beta_2, (\nu_1 + \nu_2)\beta_1]$ be output of FPHPS of type $(\mathcal{N}, (\nu_1 + \nu_2)\beta_1, \beta_1)$
- $\mathcal{N}_i = \begin{cases} \nu_2 - 1 & \text{if } 1 \leq i \leq \beta_1, \\ \nu_2 & \text{if } \beta_1 + 1 \leq i \leq \beta_1 + \beta_2 \end{cases}$
- $\mathcal{F}_i(\lambda) = (1, \lambda, \lambda^2, \dots, \lambda^{\beta_1-1}) \cdot \mathcal{G}_i(\lambda^{\beta_1})$
- $\mathcal{G}_i$ is i -th column of $\mathcal{G}(\lambda) = [I - (\sum_{i=0}^{\nu_1 + \nu_2 - 1} B_i \lambda^i)]$

Then

- β_2 of $\mathcal{P}[i, (\nu_1 + \nu_2)\beta_1]$ have positive defect: $\overline{\mathcal{P}}^{[1]}, \dots, \overline{\mathcal{P}}^{[\beta_2]}$
- $b_i = \text{rev}_{\nu_2 + 1 - \text{dct}(\overline{\mathcal{P}}^{[i]})} \left([\overline{\mathcal{P}}^{[i]}_{\beta_1 + 1} \cdots \overline{\mathcal{P}}^{[i]}_{\beta_1 + \beta_2}]^T \right)$ basis for $\mathbb{G}$
- $\deg \left(\det \left([b_1 \cdots b_{\beta_2}] \right) \right) = \beta_2(\nu_2 + 1) - \sum_{i=1}^{\beta_2} \text{dct}(\overline{\mathcal{P}}^{[i]})$

Block Wiedemann and Krylov Sequences

- f^A generates both $\{X^\top A^i Y\}_{i=0}^\infty$ and $\{A^i Y\}_{i=0}^\infty$
- $\mathbb{G}^{A,Y} \subset \mathbb{G}_X^{A,Y}$
- $F^{A,Y}$ generates $\{X^\top A^i Y\}_{i=0}^\infty$
- $\deg(\det(F_X^{A,Y})) \leq \deg(\det(F^{A,Y})) \leq N \leq n$
 where $N = \sum_{i=0}^{\beta_2-1} \deg(s_{n-i}(\lambda I - A))$
- $\deg(F_X^{A,Y}) \leq \deg(F^{A,Y}) \leq \deg(f^A) \leq n$

Block Rank Algorithm

Kaltofen-Saunders rank algorithm:

If

- $A \in \mathbb{F}^{n \times n}$ has rank $r < n$ and a nonzero $r \times r$ principal minor
- S is a finite subset of $\mathbb{F}$
- $D = \text{diag}(d_1, \dots, d_n)$ where $d_1, \dots, d_n$ are chosen uniformly and independently from S

Then,

- $r = \text{rank}(A) = \deg(f^{AD}) - 1 = \deg(f^{AD}) - \text{codeg}(f^{AD})$
with probability at least $1 - n(n-1)/(2|S|)$
- $r = \text{rank}(A) = \deg(f_u^{AD,v}) - 1 = \deg(f_u^{AD,v}) - \text{codeg}(f_u^{AD,v})$
with probability at least $1 - n(n+3)/(2|S|)$

If

- $A \in \mathbb{F}^{n \times n}$
- $\beta_2 \leq \beta_1 \leq n$
- $N = \sum_{i=0}^{\beta_2-1} \deg(s_{n-i}(\lambda I - A))$
- Entries of $X \in S^{n \times \beta_1}$ and $Y \in S^{n \times \beta_2}$ are chosen uniformly and independently from S , a finite subset of $\mathbb{F}$
- $\nu_1 = \lceil N/\beta_1 \rceil$ and $\nu_2 = \lceil N/\beta_2 \rceil$
- $H_X^{A,Y}(\nu_1, \nu_2 + 1) = \begin{bmatrix} X^\top Y & X^\top AY & \dots & X^\top A^{\nu_2} Y \\ X^\top AY & X^\top A^2 Y & \dots & X^\top A^{\nu_2+1} Y \\ \vdots & \vdots & \ddots & \vdots \\ X^\top A^{\nu_1-1} Y & X^\top A^{\nu_1} Y & \dots & X^\top A^{\nu_1+\nu_2-1} Y \end{bmatrix}$

Then, $\text{rank}(H_X^{A,Y}(\nu_1, \nu_2 + 1)) = N$ with probability at least $1 - 2N/|S| \geq 1 - 2n/|S|$.

If

- $A \in \mathbb{F}^{n \times n}$
- $\beta_2 \leq \beta_1 \leq n$
- $N = \sum_{i=0}^{\beta_2-1} \deg(s_{n-i}(\lambda I - A))$
- Entries of $X \in S^{n \times \beta_1}$ and $Y \in S^{n \times \beta_2}$ are chosen uniformly and independently from S , a finite subset of $\mathbb{F}$

Then,

- $\gamma_1 \leq \lceil N/\beta_1 \rceil \leq \lceil n/\beta_1 \rceil$ and $\gamma_2 \leq \lceil N/\beta_2 \rceil \leq \lceil n/\beta_2 \rceil$
- $s_{\beta_2-i}(F_X^{A,Y}) = s_{n-i}(\lambda I - A), \quad 0 \leq i \leq \beta_2 - 1$

with probability at least $1 - 2N/|S| \geq 1 - 2n/|S|$.

If

- $A \in \mathbb{F}^{n \times n}$ has rank r and a nonzero $r \times r$ principal minor
- $\beta_2 \leq \beta_1 \leq n$
- S is a finite subset of $\mathbb{F}$
- Entries of $X \in S^{n \times \beta_1}$ and $Y \in S^{n \times \beta_2}$ are chosen uniformly and independently from S
- $D = \text{diag}(d_1, \dots, d_n)$ where $d_1, \dots, d_n$ are chosen uniformly and independently from S

Then,

- $r = \text{rank}(A) = \deg(\det(F_X^{AD,Y})) - \text{codeg}(\det(F_X^{AD,Y}))$
- $F_X^{AD,Y}$ can be computed from $\{X^\top A^i Y\}_{i=0}^{\lceil N/\beta_1 \rceil + \lceil N/\beta_2 \rceil - 1}$

with probability at least

$$1 - (4N + r(r + 1))/(2 |S|) \geq 1 - n(n + 3)/(2 |S|).$$

Input: $A \in \mathbb{F}^{n \times n}$ with $\text{rank}(A) \leq n - 1$, S a finite subset of $\mathbb{F}$, and $\beta_2 \leq \beta_1 \leq n$

Output: $r = \text{rank}(A)$

- 1: $B_1, B_2 \leftarrow$ butterfly network preconditioners with parameters chosen uniformly and independently from S
- 2: $D \leftarrow \text{diag}(d_1, \dots, d_n)$, $d_1, \dots, d_n$ chosen uniformly and independently from S
- 3: $\tilde{A} \leftarrow B_1^\top A B_2 D$
- 4: Choose $X \in S^{n \times \beta_1}$ and $Y \in S^{n \times \beta_2}$ uniformly and independently
- 5: $\nu_1 \leftarrow \lceil n/\beta_1 \rceil$ and $\nu_2 \leftarrow \lceil n/\beta_2 \rceil$
- 6: Compute $F_X^{\tilde{A}, Y}$ from $\{X^\top A^i Y\}_{i=0}^{\nu_1 + \nu_2 - 1}$ (E.g., via the FPHPS algorithm.)
- 7: $r \leftarrow \deg(\det(F_X^{\tilde{A}, Y})) - \text{codeg}(\det(F_X^{\tilde{A}, Y}))$

Template wrapper

Template *wrapper* class `UnparametricField<double> F`

Wrap `NTL::zz_p`: `UnparametricField<NTL::zz_p>`

Partial template specialization:

```
template <> NTL::ZZ_p&
UnparametricField<NTL::ZZ_p>::inv(
    NTL::ZZ_p& x,
    const NTL::ZZ_p& y
    ) const
{ return x = NTL::inv(y); }
```


Performance of `UnparametricField`

Time in seconds to compute the N additions and multiplications using
`NTL::zz_p` directly and as the `LinBox` field
`UnparametricField<NTL::zz_p>` using the prime 32749.

N	10^7	2×10^7	3×10^7	4×10^7	5×10^7	6×10^7
Directly	4.68	9.46	14.11	18.81	23.29	27.94
<code>LinBox</code> field	5.56	11.05	16.59	22.07	27.41	32.90
<code>LinBox</code> field with <code>init</code>	15.46	31.06	46.24	61.97	76.58	91.93

`Modular<long>`

N	10^7	2×10^7	3×10^7	4×10^7	5×10^7	6×10^7
<code>Modular<long></code>	1.84	3.65	5.49	7.31	9.14	10.99

Archetypes

Three uses of archetypes:

1. To define the common object interface; i.e., specify what an explicitly designed class must have.
2. To distribute compiled code and prototype library components.
3. To control code bloat.

Used for all but the most basic objects such as floating point numbers and 10 by 10 matrices.

Not a Java interface; an archetype is an explicit class.

Field Archetype `FieldArchetype`

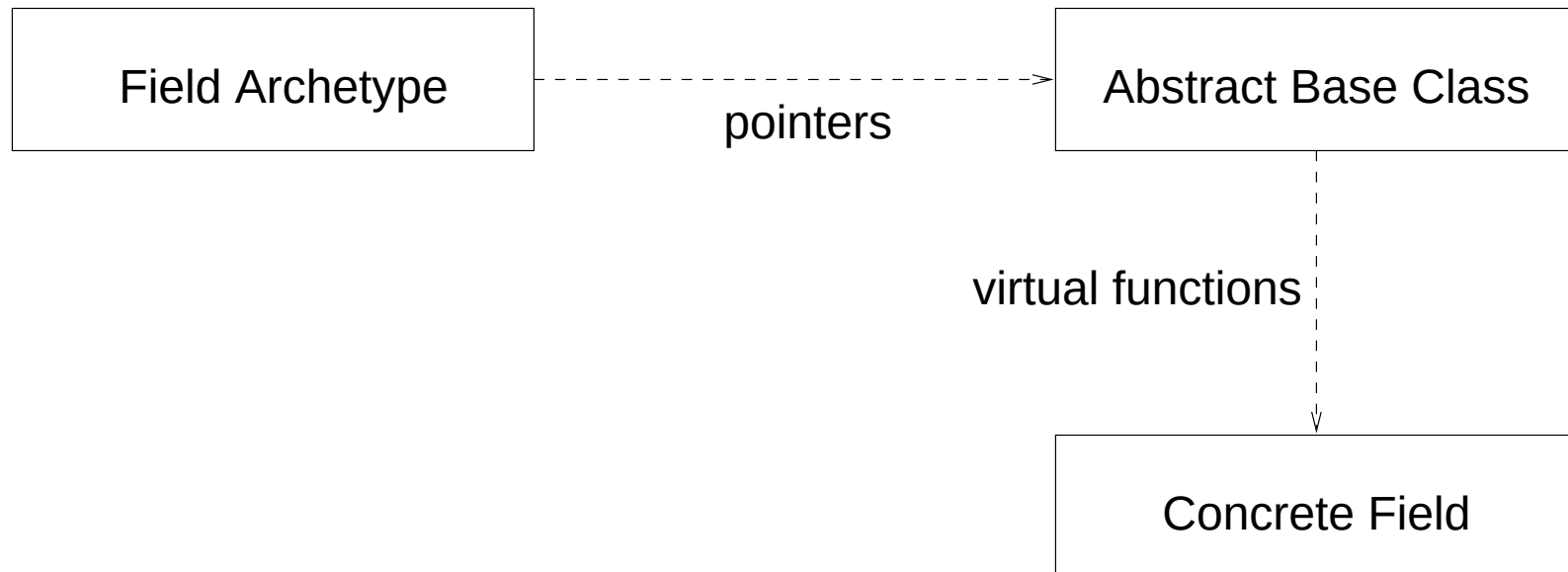
Allow C++ types such as doubles as field elements



Field elements must have (non-virtual) constructors



Field archetype $\neq$ Abstract base class



Envelope `FieldEnvelope` to use `FieldArchetype` with LinBox field

Performance of FieldArchetype

Time in seconds to compute the minimal polynomial of a $n \times n$ Trefethen matrix using the LinBox fields `Field1 = Modular<long>`, `Field2 = UnparametricField<NTL::zz_p>`, and `Field3 = UnparametricField<NTL::ZZ_p>` of integers modulo the prime 32749.

n	500	1000	1500	2000	2500	3000	3500	4000
Field1	0.91	3.85	9.09	16.53	26.51	39.43	53.85	71.66
Field2	2.93	12.67	29.94	54.41	86.78	128.65	175.6	233.81
Field3	6.34	27.24	71.87	127.69	193.69	316.91	451.80	630.95

Using `FieldArchetype`:

n	500	1000	1500	2000	2500	3000	3500	4000
Field1	1.82	7.69	18.00	32.94	52.76	78.06	108.83	153.97
Field2	3.85	16.68	39.21	71.07	114.23	169.72	232.26	323.20
Field3	7.56	32.59	90.80	153.58	274.43	440.80	626.29	864.09

Trefethen's Challenge

- $(A^{-1})_{1,1}$ where $A \in \mathbb{Z}^{20,000 \times 20,000}$

$$A_{i,j} = \begin{cases} i\text{-th prime} & \text{if } i = j, \\ 1 & \text{if } |i - j| = 2^k, \quad k = 0, 1, \dots, 14, \\ 0 & \text{otherwise.} \end{cases}$$

- $Ax = e_1 = [1 \ 0 \ 0 \ \dots \ 0]^T$

- $(A^{-1})_{1,1} = x_1$

- Cramer's rule: $(A^{-1})_{1,1} = \frac{\det(A \overset{1}{\leftarrow} e_1)}{\det(A)}$

Hensel Lifting

- Initialize

- $b^{[0]} = b = e_1$

- $\bar{x}^{[0]} = x^{[0]} = A^{-1}b^{[0]} \bmod p$

- For $j \geq 1$

- $b^{[j]} = \frac{b^{[0]} - A\bar{x}^{[j-1]}}{p^j} = \frac{b^{[j-1]} - Ax^{[j-1]}}{p}$

- $x^{[j]} = A^{-1}b^{[j]} \bmod p$

- $\bar{x}^{[j]} = \bar{x}^{[j-1]} + p^j x^{[j]} \equiv x \pmod{p^{j+1}}$

Using Minimal Polynomial

- A^{-1} is dense
- Use Wiedemann's method to compute $A^{-1}b[j] \bmod p$
- $g = \sum_{i=0}^d g[i]\lambda^i = \left(\frac{-1}{f^A[0]}\right) f^A, \quad g[0] = -1$
- $x[j] = \sum_{i=1}^d g[i]A^{i-1}b[j] \bmod p$

Contributions

- Implementation (LinBox)
 - Field, vector, and black box matrix designs (§§5.3-5.5)
 - Vector Traits (§5.4.1)
 - Subvectors (§5.4.2)
 - Examples and tests
 - * Fields: `Modular`, `Fuzzy`, `UnparametricField`, `NTL::zz_p`, `NTL::ZZ_p` (§5.3)
 - * Matrices: `SparseMatrix0`, `Diagonal`, `Hilbert`, `Compose` (§5.5)
 - * Trefethen challenge (§5.6)
 - DOC++ documentation (§5.7)
- Theoretical
 - Switching network preconditioner proofs (Thms 2.1 & 2.2)
 - Determinantal divisor preconditioners (Thms 3.2 & 3.4)
 - Equivalence of linearly generated matrix sequences (Thm 4.1)
 - Computing minimal generating matrix polynomial via FPHPS (Thm 4.10)
 - Block Wiedemann rank algorithm (Thm 4.14)